

# Proof ledger

Repository source: proof/PROOF\_LEDGER.md

## Status vocabulary

- **ESTABLISHED IN MANUSCRIPT:** a complete derivation is present in the paper, subject to ordinary mathematical review.
- **EXACT-CERTIFIED:** the finite claim is checked with integer and rational arithmetic by the standalone verifier.
- **NUMERICAL AUDIT:** a separate floating-point implementation agrees.
- **RECONSTRUCTION-CHECKED:** an expected derived artifact is rebuilt byte for byte.
- **MUTATION-TESTED:** a targeted corrupted object is rejected.
- **SOURCE-CHECKED:** bibliographic or historical wording has been compared with an identified source.
- **RELEASE-PENDING:** technically prepared but awaiting a hosted workflow, immutable tag, release assets, or explicit approval.
- **EXTERNAL REVIEW NEEDED:** no independent subject-matter review has been recorded.

## Mathematical claims

| ID | Claim                                                                | Evidence                                 | Status                                        |
|----|----------------------------------------------------------------------|------------------------------------------|-----------------------------------------------|
| M1 | Constant-width support identity and diameter-one normalization       | Paper Section 2; classical references    | ESTABLISHED IN MANUSCRIPT;<br>SOURCE-CHECKED  |
| M2 | Blaschke volume-surface identity                                     | Paper Section 2; classical references    | ESTABLISHED IN MANUSCRIPT;<br>SOURCE-CHECKED  |
| M3 | Minimum circumsphere contact balance and Jung range                  | Paper Section 2                          | ESTABLISHED IN MANUSCRIPT                     |
| M4 | Opposite-point identity and radial support inequality                | Paper Lemma 2.1                          | ESTABLISHED IN MANUSCRIPT                     |
| L1 | Transport inequality and tangent-gap master inequality               | Paper Section 3                          | ESTABLISHED IN MANUSCRIPT                     |
| L2 | Rational propagation implications                                    | Paper Section 3; exact lower-cell checks | ESTABLISHED IN MANUSCRIPT;<br>EXACT-CERTIFIED |
| L3 | Four refined cells replace 03A1 with positive margins                | Refined NPZ cells and JSON               | EXACT-CERTIFIED;<br>NUMERICAL AUDIT           |
| L4 | Complete lower partition is gap free and has unique bottleneck 03B02 | Final JSON and exact verifier            | EXACT-CERTIFIED;<br>RECONSTRUCTION-CHECKED    |
| N1 | Near-Jung four-contact deficiency budget                             | Paper Section 4                          | ESTABLISHED IN MANUSCRIPT                     |
| N2 | Affine contact coordinates and Gram formulas                         | Paper Section 4                          | ESTABLISHED IN MANUSCRIPT                     |

| ID | Claim                                                                | Evidence                                     | Status                                                               |
|----|----------------------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------|
| N3 | Determinant transfer<br>$\det H \geq 1 - S/2 - S^2/2$                | Paper Proposition 4.1                        | ESTABLISHED IN<br>MANUSCRIPT; exact<br>endpoint evaluation           |
| N4 | Dominant-edge<br>coordinate-width diagonal<br>bounds                 | Paper Section 5                              | ESTABLISHED IN<br>MANUSCRIPT; exact<br>endpoint evaluation           |
| N5 | Adjusted 98,306-ray fan is<br>feasible with positive exact<br>margin | Fan NPZ, orbit TSV,<br>exact verifier        | EXACT-CERTIFIED;<br>NUMERICAL AUDIT;<br>MUTATION-TESTED              |
| N6 | Forced-core volume lower<br>bound                                    | Exact fan volume and<br>determinant transfer | EXACT-CERTIFIED;<br>NUMERICAL AUDIT                                  |
| N7 | Three axis-specific<br>antipodal cap-pair bounds                     | Hull TSV and cap<br>certificate fields       | EXACT-CERTIFIED;<br>NUMERICAL AUDIT;<br>MUTATION-TESTED              |
| N8 | Six cap interiors are<br>pairwise disjoint                           | Paper and exact<br>disjointness fields       | ESTABLISHED IN<br>MANUSCRIPT;<br>EXACT-CERTIFIED;<br>MUTATION-TESTED |
| N9 | Near-Jung lower bound<br>exceeds theorem upper<br>enclosure          | Final JSON assembly                          | EXACT-CERTIFIED;<br>NUMERICAL AUDIT                                  |
| T1 | Universal coefficient is<br>$26220940089713\pi/2000000000000000$     | 03B02 plus strict near-Jung<br>000000ff      | EXACT-CERTIFIED;<br>RECONSTRUCTION-<br>CHECKED                       |
| T2 | Scaling gives the theorem<br>for arbitrary width $d > 0$             | Homogeneity                                  | ESTABLISHED IN<br>MANUSCRIPT                                         |

## Verification and provenance claims

| ID | Claim                                                              | Evidence                                      | Status                     |
|----|--------------------------------------------------------------------|-----------------------------------------------|----------------------------|
| V1 | Standalone verifier has no<br>private repository<br>dependency     | Path regression test and<br>source inspection | EXACT-CERTIFIED            |
| V2 | Final JSON is rebuilt byte<br>identically                          | Exact verifier and unit test                  | RECONSTRUCTION-<br>CHECKED |
| V3 | Exact TSV and CSV<br>renderings are rebuilt byte<br>identically    | Exact verifier                                | RECONSTRUCTION-<br>CHECKED |
| V4 | Binary64 implementation<br>agrees on principal<br>quantities       | Binary64 transcript                           | NUMERICAL AUDIT            |
| V5 | Base-R implementation<br>agrees on principal<br>quantities         | Named-author machine R<br>transcript          | NUMERICAL AUDIT            |
| V6 | Seven named proof-object<br>mutations are rejected                 | Mutation transcript                           | MUTATION-TESTED            |
| V7 | Paper-facing constants<br>and lower table match the<br>JSON        | Paper reconciliation script                   | RECONSTRUCTION-<br>CHECKED |
| V8 | Exact-constants appendix<br>matches the JSON                       | Generator <code>--check</code>                | RECONSTRUCTION-<br>CHECKED |
| V9 | Required release<br>documents exist and<br>preserve claim language | Release-document checker                      | RECONSTRUCTION-<br>CHECKED |

| ID  | Claim                                                   | Evidence          | Status                                       |
|-----|---------------------------------------------------------|-------------------|----------------------------------------------|
| V10 | Markdown/PDF pairs are complete                         | Pair checker      | RECONSTRUCTION-CHECKED                       |
| V11 | PDFs are readable, have embedded fonts, and expose text | Poppler preflight | Passed in construction; hosted rerun pending |

## Publication and review claims

| ID | Claim                                                                  | Status                                         |
|----|------------------------------------------------------------------------|------------------------------------------------|
| P1 | Lean ten-page Paper v1 manuscript is present                           | Completed draft                                |
| P2 | Complete release-document set is present in Markdown and PDF           | Completed in this checkpoint after local build |
| P3 | Release-candidate author list and no-affiliation statement are fixed   | Completed for <b>v1.0.0-rc1</b>                |
| P4 | Split license and release-candidate citation metadata are fixed        | Completed for <b>v1.0.0-rc1</b>                |
| P5 | Public remote, immutable tag, and release assets exist                 | RELEASE-PENDING                                |
| P6 | Hosted Python, R, and document jobs are green on the release candidate | RELEASE-PENDING                                |
| P7 | Fresh public-clone replay is recorded                                  | RELEASE-PENDING                                |
| P8 | External subject-matter reviewer has checked the theorem               | EXTERNAL REVIEW NEEDED                         |
| P9 | Peer review or journal acceptance exists                               | Not claimed                                    |

## Claim boundary

The package does not establish equality, rigidity, sharpness, a minimizing body, or Meissner extremality. It should not be described as independently verified or formally verified while **P8** remains open.

## Current bottleneck

The mathematical, finite-certificate, authorship, citation, and licensing packages are prepared for **v1.0.0-rc1**. The remaining release bottlenecks are administrative, hosted, and external:

1. public remote and release-candidate push;
2. green hosted workflow on that exact commit;
3. final named-author copy edit and page approval;
4. final release date, **v1.0.0** tag, and fresh tagged-checkout replay;
5. immutable release assets and independent hash verification;
6. explicit approval of the final README, paper, release notes, tag, and announcement; and
7. external mathematical review after release.